ActInf GuestStream #034.1 ~ Avel Guénin-Carlut ~ "Physics of Creation"

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good luck hello everybody I'm ready yes we are in it is January 23rd 2023. we're here in active inference guest stream number 34.1 with aval guadin carlu we're going to have a one hour presentation physics of creation can the free energy principle ground the construction of physical state spaces followed by about an hour discussion so if you're watching live please feel free to add questions in the live chat and we'll also have some guests so thank you AFL for joining and off to you for the presentation hello so this is a complete presentation uh I will first introduce myself in the work I am here I have a common physics I'm working creative sense under version of Andy Clark um this work is institutionally speaking the product of Kairos research which is laboratory which myself Alexi rozovski who is present here today A lot of people are funded to investigate issues of events to political through political Evolution and Community science um and the the topic will be on the physics of creation uh so creation is a atypical term let us say it will be contextualized in presentation but let us first Define that it corresponds to the construction of physical possibilities physical spaces to talk in a more technical language and the original question that motivated This research is whether or not the federal principle a piece of math that emerging the cognitive sounds which we will get to very soon is capable of representing the this process of creation so first I have tried the first run of this without getting into the technicals it did not work so I will address the technicals first we will see some Concepts in physics what they do how they work how they relate to [Music] what people typically do in physics and first I'd like to take a look at how physics explain so physics explain with laws they think that in nature you have laws and those laws are basically formal statements that constrain the possible states of a system and typically those are entirely lows that from which you can derive logically what the system will do an example of natural entailing law is Maxwell's equation of electromagnetics it is a different Maxwell than the one that is here so what you see is four bits of vectorial calculus that will tell you how two fields um later collectively the electromagnetic field evolve with regard to another so so this is a formal constraint on what e and B can do like the equality Express constraint by definition and those can be used to derive basically fully e and B given boundary conditions which is how you do things in electromagnetism uh but what is of ravens here is that any of the slope responses basically that I can Define e as a virtual field so it preciposes what is called set space which is mathematical space that represents a set of possible States before I State Delta e and Delta B is equal to this and this we have to state that e exists and he can take such and such States in that case it is a vectoral space in three dimensions um and this headspace basically it has by construction to capture any difference that make a difference about the system that is targeted for any change in E to be basically computable by math Theory it has to be represented in the state space so by construction I want this space to categorize every variable of interests in my system given this scope and measurement capabilities of my theory foreign you have underlying notion that is basically implicitly uh underlying the notion of flows and the space that is notion of symmetry I cannot get into the query detail here but I will try to be as complete as possible um we typically understand physical theories outside specifically relativity Theory and some weird Quantum stuff uh to derive from symmetries symmetries are I will say it out loud algebraic rope of transformation over the set space that do not affect the system's Dynamics as granted as its laplacian you don't have to understand what I just said what you have to understand is that there are ways in which I can transform the system which does not change what it is an example in symmetries the sailing of the Lord falama skin in trial you see that you are physically the same pattern that is repeated again and again and again and again gradually so if I uh make a small rotation uh in either direction it will not change a theme and if I repeat it and repeat it and repeat it and repeat it and repeat it and repeat it I will still change my system like formally I will apply transformation but it does not challenge basically what observable properties the system have and this is what asymmetry is it is a set of transformation that is related by a subset of smaller transformation and that does not change what your system observably is and because of this symmetries are a system that basically constrain what difference can make a difference about the system so implicitly when I from light uh low or set space I make a statement about the underlying symmetries I make the statements that basically everything that is not in the system that is not represented as part of the system is something that is symmetric that the system is Centric with regard to but on which any transformation will not affect the system trajectory at least not in a way that is relevant to my theory um it works for physics because it looks like in physics we have a few symmetries that are everywhere verified that are fundamental in some sense it doesn't work so well in life in mind because we don't have that to be specific living system of a structure I do not think it should be controversial and they have constitutive symmetries they have changes that will not affect them if I switch all of the carbon atoms in my body I scramble them between each other while keeping the relation to the rest it did not change the thing so it is very trivial in sense of symmetry but basically every that is me entails symmetries and Taylor's information that are possible that do not affect the structure and others are told affect it and this Authority is not a given like symmetries are considered to be in physics it is actively reconstructed by the activity of me if I cut myself if I fall if my bone break I will regrow basically you will have seek official tissue that will uh occur and that will make it okay again and it will change what structure is my body most importantly I I did I was not born that way like big and with a bird and with 80 kilo of meats I grew and I Grew From one single cell and this single cell just created anatomyal structure um apparently X nilo from basically genetic code and the context and not only that but life you know if itself evolves in a way that builds structures that builds symmetries and a lot of this means that you can't set a state space that is the space of life in general of any living system the Symmetry is that in the live system they change to time and they change by the own activity of this system so any living system construct set spaces and to account for it we need a theory of test-based construction this is the motivation of the physics of creation uh line of work and this is basically the I did not permit as a question it is the question that will uh that you need to keep in mind in this discussion so uh Danielle do we have any question yet okay no technical issue that I should know about okay so what we'll do is First Look at basically what we say of Creation in life in mind at the conceptual level what is the history of the concept and what they are going to do when we look into the formal specifics of how we try to recruit to represent basically recognition as inference and what limits these attempts have um and finally um we will take the reflection further and see whether we can and how far we can represent physical creation the construction of physical reality uh with the tentative formalism we have framed here so let us look at Creation in life and mind historically the term of creation it refers to autopoiesis that is that means literally self-creation and that is not attempt by an activist what became an activist so mainly uh the materana and Varina to ground the study of cognition the study of biology and the way well analytic system work they framed Orthopedic Theory as basically the idea that cognition is grounded in a way that link system constructs itself like their metabolism allow them to reconstruct a structure and the structure does not have to be conserved by metabolism so you have self-creation and this is basically the concept that try to articulate to ground cognitive science and later uh very well we and colleagues try to basically concretize this by working on how the cognition was enacted which is it is implemented in the in the body in the concrete activity of living systems and this basically um was very determinant in the history of cognitive sense and it framed the 4E approach so the extension of ignition beyond the brain let us say very very uh tentatively uh the idea that frame is that you have uh you can see the diagram that is taken from the Paulo's work and I tried to um to explain the relation between self-constitution so let us say it is a weaker lighter more specified version of the phrases and agency the idea is that you have self-constitution that is the fat living system build themselves and it is basically afforded by the way living system coupled with environments and agencies uh lighter higher order of regulation of this disability to define basically the terms of your coupling with the environments the ID generates is that living systems there are systems that have a structural identity that they actively generate and sustain under precarious circumstances there is no benevolent universe that tells lives that help life be life life has basically to generate environment where it is life and in doing so it generates and it sustains what it is its identity and all of this is necessary for sense making so let us say cognition to be something to occur this is in my opinion and in to some extent in the power of our opinion somewhat were deep it is a lot of Concepts a lot of Greek words and people have tried to frame this in a way that is more mathematical and formal the question it generates is how to formalize these self-creative property of biological organization and cognition and in a way that is basically grounded in a material information flow in things that are concrete physical structure we have a pretty good answer to that actually we have a pretty good work that made a point that you had related this somewhat high-level concept to basic physical processes this work is the work of Moreno mosio and monteville mainly who work in a forgot the name but uh Institute eventually in the pay Basque in Spain and the idea is that living systems are constituted by a closure of constraints so the constraint is physical thing that shape how other physical things work very simple if I have air in a box that is closed the air count with the box it is constraint okay and the property that individuates living things is the fact that they are they are considered by a set of constraints that canalize metabolism in a way that reconstructs the same way the same of constraints and so uh from there you have self-constitution you have also um structural let us say notion of biology that does not depend on the matter it is made of but on the way it is sustained that is I think pretty good if you want to have a physicalist notion of life we have a very basic ontology in which constraints they shape metabolism and just shape information flow as in the way uh living stem rug to things and that grounds the autonomy of living things the fact that they can recreate themselves basically but nothing gives a clear link between this and cognitive meaning or anything we could recognize as cognition so this is what we will try to articulate next so how driven systems as we just defined can create meaning the notion I would like to import is the notion of active inference which I will not get into the mathematical detail here but the idea is that the way the flow the dynamical flow literally that is untailed by an agent and its environment it individuates states that are being full basically an agent will predict in a relatively familiar sense um Sensations that it will I don't always say that not perceived Percy but undergo and actions it will Undertake and these predictions are basically what's underlying meaning for the agent it underlies what states will just swap into shape its own action and be meaningful to them Um this can be framed by in a pretty a simple poetic move uh by setting that biological interactive systems they do not simply infer what is they bring about uh their own reality by actively inferring what is Meaningful to them by actively looking for states that they think are will even fault their own structure and then you start to have a stronger relation between uh what a physical system of constraint is and how it invigorates states that are meaningful but you still don't have a formal theory of basically how this metabolic flow builds uh learning or creation like builds a new states of possible a new space Sorry of possible States that is not the one it began with uh I had I have a description of a structure that is the set of constraints and this produces basically an activity of the system a flow over the structure and this flow is somehow constrains or shape the structure the system takes at time t plus one which itself produce a new function a new flow etc etc etc so it is this Dynamic of um I don't want to call that cell for functional unfolding whatever the the way uh a system of constraints produce a flow that three produces the same constraints that we want to formalize um the question we could reframe as such and we could reframe as how does cognition the ability to understand integrates information about the world it can acts within biological organization because cognition is basically the flow of political organization but this isn't held by it everything that I see is something that I but to say bring about that I look at because of the biological constraints I am and therefore the question of creation can be reframed as how the meaning I project into the world is brought back to shape my my own structure and this is what we will try to frame mathematically um in the framework of active inference uh Daniel any questions so the attempt that we make here is to see if we can how basically we can represent creation uh the process of inference which is likely the most minimal highly recognizable way a quick frame cognition as something that is distinct from Simply it's much of an implementation uh let us look at a piece of math that is called The Foundry principle and that is the formal motivation of rounding for achieving friends it will be the hard part for people who do not do maths or people who are not already familiar with uh principle I will work with slowly which is a piece of math that frames statistical in France from the medical System Theory chemical System Theory is basically the idea that you have States and they move as a function of each other and what the fep does is to prove that if I have something that is called a Markov blankets so boundary that's any information must travel through for two subsystem to communicate practically then you have a synchronization that emerged between states that are outside this boundary and states that are inside this boundary and not only this but if I measure the every state that is on the inside of the boundary entails belief and titles equal distribution over the external States and the flow of internal States is such that it minimizes the international distance between the actual Excel States well so what I can observe and what I believe to be the case where I of course is the internal systems so that was abstract the basic unformal meaning of this is that if I have a chemical system and I have a boundary that acts to mediate information exchange between two supports then both the parts can basically represent each other and hold belief in each other in a way that is constructed by the systematic synchronization across the boundary it is uh quite abstract and quite minimal but it is a foul proof that we can have representation kind of cognitive meaning uh from a very very minimal set of hypotheses which is basically a consistent causal structure and it is a pretty good case that uh basic technical system for example staff constraints constraint will emerging robots this kind of very basic non-cognitive things can entail cognition it has been argued that it does underlie agency and creativity and similar Concepts that wouldn't tell in my opinion creation as a prerequisite so I will try to see whether the fep actually can represent creation as the ability for agent to individuate states that are spinning to them the thing is that the existence of work of bronchitis is monstrated for any complex stochastic chemical system so chemical systems that are random and this transmission realized as basically something that moves in function of itself Plus random noise it is a variable formulation I really any thing that exists can be framed as such so we should be optimistic for the ability of this our reason to represent the cognitive the production of creative meaning by basic biological system and therefore creation which is another case because when I frame uh something as X of t equal something something well I have to purify next we have to define space of states that is X I have defined creation as the ability to construct basically physical possibilities so by construction if you frame something within the medical System Theory you have to individual states a priority what you have written down is not Croatian it can be a grunt operation can be interesting in the city of creation it is not creation it is not the construction of physical spaces and it is likely not the construction of cognitive meaning at least in the sense that an activist and Auto poetic Theory people minutes Justin writes what become measurable for a given subsystem within the stochastic chemical system which is a lot that is not again creation let us think about how we should go we should think of the preparation there will be more math here I'm afraid but the first system is to observe another system it will have to lose information about its broader environment it is uh pretty easy well it is a something can derive from first principle uh because basically um if you frame it as a Quantum system observation will entail entanglements by definition I observe a system and it becomes something that tells in the States because I become entangled with it so it started to become physically synchronized with my States but I cannot synchronize with everything all the time so if I act so as to have more knowledge about the object I act so as to have less knowledge about my environment so there is no basically lossless observation it is something that can be derived in the content setting that can be reframed in other way in classical settings because of the way observation will either induce heatings information loss or cause myself to forget information you can pretty much derive it in a variety of case it is likely true in all cases and it means that observation as it is constrained by by on Flow and the Y couple of the environment it does not simply entails the observation of pre-existing while invigorated physical States it brings about well the integrating the well-defined survivals for me to observe so the way how to say and given this it is hard maybe not possible to frame a theory of from a mathematical theory that entails a very well integrated physical States if I frame such a theory of observation or measurement of cognitive meaning it will not be General and I pretty much have to work creation on the on the very fundamental scales at which my theory operates and this is something that is arguably done by uh reframing of The Foundry principle in for generic Quantum systems by mainly Chris fields the idea is to use um category theoretic equivalence between lots of things I will definitely not get into that to reframe the synchronization Dynamic that's in view that builds cognitive meaning according to the Friendship principle has a series of binary measurements are equivalently symmetry bracking operators so the way a chemical system synchronized with its environments under the Feb is reframed here as basically a series of binary questions that I ask my environment and that enforce my interaction with it in such a way that the answer to my questions become physical facts which you can derive from a rather General Quantum information Theory background but that gives basically no structure whatsoever to the measurement nothing in this series says what kind of system can ask what kind of questions and this means that okay this drives creation maybe drives the situation of physical possibilities physical observables but it does not constrain at all what discretion is so it doesn't explain much and doesn't help much to understand how a living system will create itself um so we still have somehow to recruit uh the cognitive and the medical formulation of the fep that I discussed very tentatively earlier so let us take things from there um I have turned uh during this formal discussion from the notion that we have to represent Creation in the living and cognitive world to the notion that somehow it is necessary it was on creation of physical reality like the underlying states of physics this third part with this third part will serve mainly to discuss this claim and what it means and what we should do with it particularly first um I want to talk basically the decline there is no a priority physical where it is a pretty basic claim it should not be treated as exceptional it is necessary property of a coherent physics it is necessary require property actually of the formulation or description of a world that is beyond what we call metaphysics let us be more specific in the argumentation I call these argument from ontological consistency anontology that is naturalistic that serves to present nature if it is consistent cannot call onto external object objects that will not be natural to explain the properties of its own objects for example if I tell you um this thing is black because it reflects uh the light in such and such way and I can also explain the properties of light it is self-consistent if I tell you this thing is black because God will eat and I cannot explain you what God is physically it is not self-consistent there is no the property that is framed by Alexi rozanski who is present here as the property of self-functional consistency that is that the Anthology should basically include itself um the discourse we have on physics is unless we are very radical position about the relation between human and the other world it is something that exists in nature and that is something that we need to account as an element of nature these entail two things this one's one thing actually it entails that uh scientific presentations the models and Theory we have and the property of those model and Theory have they must be accounted for As Natural entities in their own rights and there they must be at least in principle neighbors by the tools of physics and science and these entails in turn that I cannot have such a thing as a law asymmetry or a set space that pre-exist physics any statement I make about physics must itself be explained by physical processes I will elaborate on what it means uh the view of physicist is typically that real theories and those theories are representation of the world that just so happen to account for fundamental structure of the world for example we have the Maxwell law of electoral magnetics electromagnetism and those laws are representations of a force that exists and that is directed by um as you want I think by a symmetry in the hamiltonian sorry um asymmetry in the quantum property of systems but this symmetry practiced basically any attempt to represent an any physical process it is a fundamental property what I say is that you cannot have a fundamental property you can't have something that predates physics and cannot ultimately be explained and if you admit such a thing your ontology is not consistent because you have to admit some things that are beyond that predate nature to explain things in nature which is the opposite of what self-consistency is if I want to frame a world where you do not have explanation that call unto things that are Beyond physics you have to have lost symmetries in the species that are constructed by physical and it is not a promatic view to have at least epistemologically it is necessary let us discuss what it means concretely we have cognitive systems dot creative systems they perceive and this perception brings about collectively relevant states which the cognitive agent experience as real we have physical observers this physical observers bring about what we call punter states that are well invigorated States that are miserable and those States constitute physical reality at least as any given Observer can experience it what I say here is that this is not epistemic Notions the important States and the cognitively relevant states that perception The cognitively Driven State under perception by agents they are real and they are invigorated as real by their and if we frame it properly the qivp at least what we could do with the qivp it represents how protamines at least particles agents that have properties that we classically understand as mindful they individuate physical observable which means that they create the species which means that this is a theory of creation not simply a theory of how access reality but a theory of how we get reality in the first place to do anything like grounding properly this claim it will take a lot more details I will try to work by showing what the search and Theory would do because obviously I do not have such a theory that is frame formally from start to finish as of now and if I did I will not present it now let us look at the way constraint basically creates the relative experience this is a diagram of how as human management from the world the social world basically we understand things like constraints over what we do and can do and what it entails I understand that if I talk to you in proper English you will understand if I talk to you in French Australia you will not understand understand that if I address it will be inappropriate and if I hold myself in a certain way it will lead you to believe me more or less understand that if I start speaking very loud it will disturb people and they will act for me to stop talking very loud I understand very basic facts about how this world work and I understand how this constrain the action I can take and the consequences of that action what experiences basically a social cultural landscape which exists uh independently from me and that I act Within what I claim is that this social cultural landscape evolves in a way that is creative that is basically abuse new Norms exist that did not exist 10 years ago and that is even more true for one million ago or before human worth seeing and the way I can represent how I don't tried to understand this social landscape and possibly fail or erode norms and the landscape in such way it is equivalent to representation of the landscape itself and the constraints that constitute this landscape and the way the flow of landscape reconfigures the landscape so what I did here is to basically Prime theorem of Creative Evolution this of a set of constraints well of a structural landscape uh on the way individual agents act to understand and act within this landscape I have a sense in which constant observation and cognition about physical here a social reality but still physical in a sense that it exists in a physical how this landscape can be created by something that is cognitive and this basic ontology if I am correct here is something we can mathematize I hope so and that we can frame as a basic theory of well the individuation of new physical possibilities ER creation ie uh credit to this graph is given to nabernacle what is of importance here is The Duality between two representation uh two perspective the perspective of agent that basically enact circular social cultural constraints that the experience has a physical fact as a reality they cannot go beyond or Escape or maybe a fact in Brute Force that's possibility but not Escape and the perspective of a set of constraints that indigenously unfolds in a way that is creative and that is a formal intuition I plan to build on and I can say something a bit more specific about how it works by analogy to differentiate principle and the allographic principle basically a creation operator in my opinion I could argue for that I will not do this in detail right now it's something that goes for a space of it's not actually a space of physical possibilities it's a space of potentialities it does not have possible States and impossible states it is just the statement that something is possible it is very close to what Greeks call chaos so you have such a space of possibilities and from the space unfolds a pair of an individual observable which is called efr environment and an agent that is capable to observe this survival and the agent is also capable to maintain uh boundary between itself and the observable or basically types available to stay existing as I'm survival to be observed and from this I think that some mathematical operator that does this basic flow is capable to describe the construction of physical spaces as they are the unfolding uh strictly strictly no metaphorically a physical structure as basically the yes the emergence the co-construction the integration as you will of uh well invigorated observable States and observer that is capable to see the state and biasing these states maintaining whatever worked as an integrated space of possibilities before as a well-integrated boundary between itself and the system let us think for example of a piece of paper I an agent can basically write things on it and by writing things on it I create new physical apps available which is the content of what I've written and the facts I write on it and or read it makes a piece of paper not a piece of paper but a boundary between myself and a semiotic space where the writing occurs this is the basic Dynamic that's must be framed for us to understand cognitive meaning and that if we framed in such way could work to describe the construction of the species X nilo because literally you do not need physical possibilities to be a thing for B to exist it is the point of this presentation you don't need a space of possibilities to pre-exist the in the duration of the agent and environment this is why it is relevant and this is why it is hard to formulate and given an operator with this property we can pretty much work it back on the foundation of physics and see what happens so have a physics of creation let us tie the threads of this presentation I will formulate three take-home messages we have something that is called the fep that is now framed that is mainly framed as a Bayesian mechanics as informational enrichment of the mucosystem theory but that can also be framed as something in Quantum information Theory it says that agents actively infer states of meaning through their engagement with the environment in a way that maintains their structure this gives us a formal background the quantum frame it on the sorry Quantum framing of its gives us a very very tentative and hard to say I would not say ill-specified but very few specified it's it does not does not have a lot of detail so it gives us a very not detailed background uh for the study of how cognitive meaning um underwrites the construction of physical spaces but because it is very attract and not very specified we have related in a much stronger way to the classical formulation of the FTP where the internal flow of the agent and the synchronization manifolds that emerge from the interaction of a very specific topology of a specific very specific geometry and can be equipped with meaning about the underlying physical structure second we have to admit that past materialism is a thing we have to admit that there is something wrong in how we view science let us uh quote Einstein if without in any white disability system we can predict with certainty the value of a physical quantity then there exists and the amount of reality corresponding to that quantity by this account nothing is real which is a very big problem a very bad position to hold and we have in some sense to account a faulty role observation in creating Well mutating Physical States not only for quantum physics not only for cognitive science but for everything in which is in some sense entail by quantum physics in the same some sense and tail by cognitive sense so I don't think it is a very big deal to say that but I think it will be taken as such the statement I want to make is that not only doing away with this brand of realism is compatible with naturalistic damage with science but it's also necessary because for naturalistic science to be a thing I have to represent myself representing as there is something metaphysical that comes to play is there as attacking myself as a detached non-physical describer of the world or by postulating some kind of god-like pre-existing structure that I just happened to stumbled upon you will have something that is beyond physics in some sense if you don't do away with physical with this brand of centrifugalism and um it has been said to me that it is not a new view no it is not it has been framed as a criticism of the national Fatman thing in itself by a Liberties philosopher about three million years ago the fact is not new is not really a problem in my opinion we should go with things that seem true whether or not they are exciting and novel and the fact that we can have a cosmology without uh practice in nature and other people do not find any problem with it should be a sign that maybe our insistence that no there are things in themselves is not well motivated and finally what I propose here beyond the physical considerations and a discussion of what some piece of math do is very very very very basic not draft but drawing of physical creation that holds in two movements the first is abstractly the invitation of something from nothing of all the word cows um given a space of undefined possibilities B uh you have in iteration of an observable that is well defined and of an agent whose observation of e also invigorates a boundary so we have this kind of framing codes work as a handle on how to write the space Construction it does have a very strong category 30 called filler from the way basically the content of B is not what's at stake this diagram is pretty much agnostic concerning what B is and is not and it must be because it underwrites the becoming of the and there is a lot of work if I want the fixed operation to be a thing and framing properly what kind of mathematical operator basically follow the flow of this diagram and what those mathematical operators do and the kind of physics like Candlelight but for this I have a constraint case I have decays of how humans collaborate to build and recreate social norms because it is a case of creation like it is pretty much the single case of creation I could convince anyone is a thing there is money because we believe there is money and it works as we believe it works because we believe it works this way it is very true statement to say that Norms constrain us because we believe in norms and The Duality it affords between the representation of single agents trying to understand and act within a social world and the flow the increasing flow that is created by the social constraints and that recreate social constraints they just practice specific it's much more specific than the vague operator I haven't left and I know this operator must represent that so it gives basically some level of grip at this entity level on how the fixed up creation would work and they would work by describing the unfolding of physical structure as the construction within physics of meaning and or constraints over what is observable by vivanagerty Observers and uh yeah pretty much we have to work from there because I do not think there is there is no self-consistent in the sense I introduced earlier there is no self-consistent Theory of physics if we don't do that so let's do that um this work again is uh to some extent my work to some extent the collective work of keros which is basically it is past particular battery that works towards uh an active sense and acting naturalism a sense that his naturalist in the sense that it plays The Observer within the world and works within well-defined causal ontology rather than less well-defined laws and Abstract principles and it tries to do so in a way that enables people was in humans to understand their reality to basically use this knowledge about how the world Works to constrain their own behavior and navigate the world especially given well the recommendation of straw constraints that will we expect and be at play in the coming decades so I strongly suggest you take a look to our website and our general line of work and consider if you want to talk to us or join us the QR code is I believe a link to our website so uh thanks to Maxwell thanks to Danielle thanks to Alexi thanks to Jonah thanks to pretty much everyone who helped me write this and think it through it was big work uh so I will run through this slide if someone wants to read them in the video and now we can go uh toward the excellent thank you thank you Alexian Maxwell welcome back if you want to join and please either of you provide a first statement or any set of questions however you want to continue just go for it you know what give me five minutes I will uh try to um destroy my heads so shots but don't expect me to well do you want to share your perspective on the project things that Avo would be familiar with while you're here uh yeah uh I Can Begin I guess um yeah so I worked on the metaphysical ontological path and yeah to me I will give it a good uh good presentation yeah it was really nice to see it's framed into the horror way of talking about everything like linking it to FTP and then to other stuff so yeah the um so the point of the if I can share a bit more about uh the metaphysics of the ontology um the idea was to create a an ontology that is naturalist and as fep seems to be really naturalist then I wanted like to make it work together at first it was only in my own my own field that I developed this Anthology and then are very said to me that it was really linked together so that's what what we try to do here yeah what does that look like how are you evaluating what a metaphysic is or how you know when you found the right one or an incompatible one yeah so the basic question was about the question of what exists more generally so that was my my anger of attack if it's a thing in English yeah so um there are many discussions about what there is in philosophy mainly in the 20th century due to Crime I think it is popular in general in the culture of science um so I followed these debates on what there is and pretty much there are a lot of views about what there is and what there is not so I was a bit how to say that disturbed by the fact that there are things that are um and I wanted to give it a try to try to talk about what seems not to be and to make it exist in some way and so um how to say that better like generally what doesn't exist is linked to Consciousness like Pegasus or let's say stupid sentences such as the round square copula of the clay College uh something else such as I don't know some weird not really mathematical entities and I was like okay so uh I agree that they do not work but at least we are talking about them so can't we just say they exist in some way and then I went into social ontology uh and I found that we can call them social artifacts and then I was a bit relieved with this view so I tried to to make it work with also phenomenality or cognitive or personal experience Human Experience because being a social artifact doesn't make it all so yeah then I I just tried to make it work between ontology social Anthology and cognitive yeah that's it and also yeah so uh something else Avail talked about the self-referential inconsistency so it's a debate uh uh I've found by by working on ontology and I was really into it the fact that a theory is inconsistent if it doesn't apply to itself and and I found some articles about it which were against naturalism that's one kind of naturalism only so it was mostly uh eliminationism and I saw that the people who were working on that they didn't know about a lot of naturalism and as I I work with Kairos and in my history uh I found that there are many kinds of natural disease so I saw that the the argument doesn't work with other kinds of naturalism such as the the one that is developed here yeah I hope it's clear yep I'll ask one question on this topic from the chat and then feel free to take it in any direction or Maxwell So Adam rostowski in the chat asked thanks Apple question maybe for Alexi what if State space is considered theoretical ideology rather than causal ontology it's presupposition would then not be a threat to ontological so what if State space is considered ontology maybe just unpack the question and then give a thought yeah I will write it now in order to reflect on it perhaps I will just a direct question what are the state spaces that you're concerned with how would you characterize what they are by construction test spaces there are spaces of postulated physical the other particular I do not understand what the inertia of Neurology how it works here you have it's pretty trivial that what we usually call ideology will Define what a person or political system or institution is capable slash interested uh in seeing it is um it is a topic of Scott single States foundational let's say work of federal case anthropology that books at basically how whenever States try to do things uh are very clearly usually for the population their main concern in effect is to build the capability of control and usually they do so by cutting through complexity aggressively a very basic instance of that is scientific forestry that is I can say it is indeed for a stream um the idea was that partially did war robots loads of them we also needed to be able to predict how many more modes more boats they would build on a given time frame so I need to know how many compatible trees they had but you don't have robot compatible trees and not worry about multiple trees you have trees or different size different ages um different sounds so if you just continue trees even with a well-defined measure you would not really know how many trees you have so they just raised for rest and planted homogeneous patch of given species at a given time with even soil so they knew that once I knew they had a 64 uh I don't know what trees there are I actually know what trees they are I don't know it's totally in English so when they had a I will it's not Oak but I will say Oak to say something uh when they knew they had a 64 uh 30 years old well that's 64. 30 I did say 30. so that's a 30 30 is Oaks and that's it they can cut the trees and they know exactly what they can do with it um so this is a way in which what something we could account as ideology so the notion that we should be able to control blah blah blah blah it's um basically construct physical reality in very concrete sense because the country like plant other trees that fit their expectation it is a form of Niche construction it is a form of um how you call that cognitive accession in the sense that the loads like what needs to account for our trees are they just upload it onto the environment by making trees actually all the same and it is a form of meaning projection by other stuff winning which is that they need to count trees that are barboats compatible and they project it onto by how actual forests are organized but and this is uh these those are physical possibilities as much better State those are set spaces we have a state space that was very complex with a lot of possible trees possible species possible size that is a um brought down to how many trees I have it is uh could you repeat the question word for word please because it was not clear to me how audition of physiology intervened the causal yes okay it precedes a space it is the ontology of uh creation slash constraints that I have um waved that um somewhat impressively at the moment uh set space uh R I think I think I will be able to demonstrate it but I counted the moment they express the point of view of an agent so I imagine that this is what is meant to be introduced by uh a logical Theory theoretical ideology and um I do not think State space are a threat to ontology I think ontologies that consider stat spaces as fundamental are float ontology is fine thank you Avo Alexi feel free to chime in and then Maxwell if you want to add anything um so I think I will join available on the documentary um also I would like to say that um if we take quiet division of ideology if I will not explain this now that well yeah so the idea I think is that the concepts are Auto and not fundamental stuff that's what I think it is implied by ideology here um so what I'd say is that you give the value of being an ideology a consolantology depending of your your goal and your criteria but at least minimally you're the stuff you're talking about it is that's the the fundamental stuff uh that are that I would say yeah so even though it is ideological even though it is instrumental let's say minimality is something and it has consequences on the world and that's what I think I've ever said with the uh nicheconstruction of social time it is didn't really say that in these words but yeah even though your IDs are let's say stupid or upset and the and people appear to hear you then your ideas they are in the world and they do something I hope I insert like if not then I can see the questions Maxwell do you want to add something or I don't currently have anything burning to say uh I I will I think this is the uh well this is the nth iteration of our discussions around this um I mean as always I think you did a great job presenting what I think is uh you know a genuine issue uh and I mean I I've just been rehearsing the same um you know appeals to multi-scalarity um I mean I I find it interesting uh that you uh would speak uh an answer to the these uh issues in Quantum fep um precisely because of its scale-free nature so you know you can't appeal to a space-time background when speaking in uh the quantum fep formulation so I think uh that in my mind that makes things um potentially more difficult I see why you would want to do that because of the focus on um you know the the definition of an observation as a physical interaction I guess I'm my uh my kind of uh overarching question is whether the multi-scalarity of classical fep helps at all uh in the sense that we described uh before so you have a set of observables some of them change too fast to really like disclose or allow us to track reliably some phenomenon behind the blanket others do in the latter case we talked about states in the former we talk about random fluctuations so I suspect that you know the the kind of definitional uh uh discussions like uh by which I mean like you know definitions of States like these kind of foundational issues that arise uh In classical fep might be helpful hereabouts uh I also suspect that you'll disagree with me so I'm looking forward to hearing your thoughts so um basically uh you have a the distinction is clear between us at least as you friendly problem you seem to be realist about the physical possibilities about spaces which I'm not I have another month for non-realism which is a self control what ontological consistency and determinants of self-constituency and selfless resistancy that brings us to say that you can't have preemptive spaces in which uh to frame in which to classify have physical observables that somehow pre-exist physics nature the process that are within it so that is one theoretical arguments the question is then can I demonstrate conclusively that you do not have physical survivals before observation but State spaces do in fact reduce the perspective of server and basically if I can demonstrate it the the debate is over until then you have you can basically uh hold you can hold structuralism but you have to account uh of how thing exists if they are not fundamental I think uh multiscal fpp can do that so it's not a critical problem for you everything holds on the formal demonstration that we do not in fact have uh set spaces as a prompted physical effects well I'm just I wouldn't push a little bit back against your realism versus anti-realism uh distinction here I'm really just talking about uh well I'm really just talking about like at the level of the formalism you start with some of the observables which seems to be what you are also uh staying and then you uh can talk about the way that those observables are related to uh some other things which may be latent all of that is a formal statement as far as I'm concerned and like it is useful whether or not it maps in some kind of fundamental realist ontology sense I I guess like if if there is a there is a kind of transcendental overtone to what I'm saying in the sense that I I you know we've argued in print in this on the map territory fallacy fallacy paper that the um the fep theoretic maths and you know the related cluster of principles unitarity maximum entropy and so forth provide ultimate constraints on any model building exercise just full stop so like again that that doesn't presuppose any realism about what maps map onto it's really just a statement about like the construction and maybe the process of constructing you're corrected since I wasn't it was uh attributing I I don't know how to say it's English it was not accurate to frame you as a realist about spaces it was correct however to frame you as an instrumentalist which I refuse to be uh because of naturalism basically I consider it necessary to frame causal ontologies that map onto reality properly as fine grantly as we can and to do that we have to include basically the Constitutional relation and this is when everything becomes complicated because you can't say if there are such physical possibilities then this occurs you can't use that language because by doing that you frame physical possibilities as something that is abstracted from the underlying physical reality you um decouple the formal from the actual I think whatever ontology we use must be virtualities but must corresponds sexually to what drives reality so I have a it is the contrary it is a i who is the realist but the constraint I was going to say maybe uh maybe actually the so I guess you know basically this clarifies I go for ontology that is consistent in the since I friend and the constraint over that are much much much much stronger than the constraint of our instrumentalists ontology that just yeah we say interesting things which is not actually easy it's just much more easy but suppose we started from the assumption that we don't have to have a strong position on the reality of State spaces uh does that affect your argument at all so you know if I were to say like I I just as the practitioner of science start from my observables you can be a radical phenomenologist from this point of view if you want you're just starting from your observables and then working out like what resists your manipulations or interactions um yeah does that in your View kind of allow us to move around asking the questions you can move around with oh about the fix of creation no um move around yes understand some things with creative meaning yes you have a fep grounds cognitive meaning in clinical center it's huge deal what's it what it does not is afford the ontological consistent Theory of physics and the difference between us is not what we hold the fep to do I think we are both instrumentalists you as a realist the difference is that I see an affordance to build a much stronger self-consistent Physical Therapy that draws elements and the general Philosophy from the fvp the philosophy of practice in France basically excuse me I push in this direction because I think the goal is variable it is ultimately an ethical call if you do not agree that the goal is variable you should not agree that it is useful to push it I do not have ultimately unsure that so maybe uh there is a kind of Conti intake um here because I I I mean I think in the at least in the history of philosophy this uh split between realism and anti-realism or instrumentalism I mean uh like I think it's perhaps overly simplistic as a description of the potential options uh there you know there is a whole kantian line of thinking uh that talks about uh you know something intermediate in some sense um so you know yeah I don't know uh just a thought so um I couldn't liberate is uh Adam in the room a damn if you're in the room uh Houston shots I agree that not in the room but I can read another question whenever you're done I'll read another question from the chat yeah but I was going to say something which is that I agree that the basically is the distinction between her instrumentalism and unrealism must be dissolved because both none of them are liturgically consistent both suppose representation relation that is outside the world and the question is whether the words as a formal construct correspond to the reality as a structural thing and there's not a concerned question to ask the all of the things are in the world we have to have a theory of how things are true or not that are false for the statements we have about the world to be in the world because they are and so from once you agree that uh ontological constances something to have a lot of the distinction we have in Western philosophy will be dissolved which makes communication complicated of course Maxwell you want a closing thought on that or I'll ask the next question um you can ask the next question I think that that brings us uh nicely to uh yeah this is from

Alex Kiefer so Alex wrote I love the positive story in this paper insofar as I understand it which I'm still working on and the idea of treating even such things as state spaces and scientific reps as amenable to physical descriptions but to bounce off what Maxwell is saying I'm not sure I see the Deep critique of thing in itself realism here probably depends what one means by that I tend to think of the strictly incoherent kantian idea of something unknowable behind phenomena so why is treating state-space creation as part of phenomena at odds with that just to pick up on that I think um you know we you don't need to be a strong realist uh you can just believe that you know you are more or less getting it right and the the semantics that we've proposed uh you know that uses active inference and the free energy principle is a kind of uh formal basis but precisely uses the fact that you know you have a certain perspective you have certain beliefs and your beliefs are going to be more or less wrong uh and that the the prediction error or the free energy is precisely what like what you need to uh the kind of minimal ingredient that you need to hook you into a situation enough to uh you know be doing something you know pragmatic that you might call truth-telling um I think the uh yeah that that's really all you need from a formal point of view is just the ability to be wrong um and to change your mind in light of new evidence so my claim is not that everything in physics is false and um formal statement about postulated possibilities of no epistemic value or trivial epistemic value my claim is that those representation as any representation are um relational they write two objects and the truth value or value in a sense that we have to give the meaning we have to give to that if we are to give anything at all it must be grounded in the study of physical Dynamics that's underlying their margins if we are to have again an ontologic consistent Theory we could just say that sense is this course that is not entirely false and say things that are not entirely false I could do that again the notion that series physical this course must be anthologically consistent is ultimately a sequel like it is a segment on what you should and should not do I can say that it is better because x y z you can say that XYZ are not meaningful targets to have and that whatever we do is okay essentially or maybe it's better than a XYZ uh but the notion that I am I do not critiques that spaces but at least it's an open point I have to find as a Critic to for it to resonate but the point is that species are rational property and that to study this uh Russian rationality it reframes the entirety of physics its epistemology it's ontology it's cosmology and if you look at the flow of this Theory it's not pure obstruction eternalize social Evolution so evolution is Agent predicting meaning as norms and then the normal swing that things it is something that is very basic for us to understand how solar system evolve and yet there is no such Theory there is no theory of social Evolution that do not presuppose this variation and Abstract away and in form of structure and Metabolism it simply does not exist so the non-existence of naturalistic theory of revolution is maybe a pretty big problem for a purely predictive uh pragmatic value standpoint from a more abstracting point we don't have a current cosmology we don't have an explanation for why why there are things and why the things that are look like what they are so physically we just have to postulate that stats by space time is such and such for physics of Excellence which is a problem we can't have a physical therapy that is that stands only on the fact that its statement does not give rise to obvious problems we have to have something beyond the statement we have to have some way to account for how this statement became true and it is a pretty basic point to make in Psychology biology sociology we can just say things are such and such we have to say how they become such and such and why they become subject search which is uh nothing in evolution nothing biology makes like nothing in biology makes sense accepting light solution and it is because those are systems that are I don't like theological let's say leonami color those are still not organized that are coherent that work in a specific way and so we have to cannot just say they do X we have to say how and why they do X and this epistemology has not has not permeated physics basically because of the belief of physicists in a platonic truth of symmetries or formalism or symmetric uh I have a question uh for you uh well so I'm thinking about this in terms of the direction of travel of our research on the free energy principle uh which I mean there are two main components to uh that we could discuss that are relevant here the first is a move from state-based to uh path based or past integral formulations where what what what's at stake is precisely probability density functions not over individual states but over trajectories entire trajectories of the system where like what what your generative model or lagrangian is describing is the probability of some specific history um so this kind of yeah intrinsically historical formalization um and on the other hand uh as a kind of second uh main direction of travel uh really formalizing these kind of three nested scales of inference where you have uh inference over uh State factors and individual states like you know color and red green blue and so on um inference over the parameters uh of the model uh the characterize the relations between uh States and configurations of states and finally the structure of the model which is really like the uh the actual uh you know relations and functionality of like what is connected to what um so one of the things that we're uh you know working on is getting to uh the mathematical foundations of these uh basically mutually constraining scales of inference uh so suppose you know you had access to a path based description of the free energy uh you know formalism uh supplemented by a clear uh you know a clear formulation of inference uh nested within a parameter optimization nested within structure optimization you know the so in other words like assuming that we solve structure learning which is a big assuming but you know if you granted those two things would there be still much of a problem you think from a formal point of view if you could just like move in the space of all possible model structures and paths through such spaces would you not basically address what you have in mind here um yeah three things first integral it solves the issue of preset meaning for biological systems and the issue of well use of that already but it is not something like code on here so it would not help with the issue I've discussed secondly structural learning the question is how basically living things project structure if we have that we would progress in a big way uh Forest area of cognitive meaning and agency you will still operate within a given space of possibilities you will have a attraction slash contraction of the state of possibles by the migrantry of what the fep is it is it builds uh manifold so the manifold like the thing will evolve on the manifold where it will evolve um so some of this that space will become trivial some of the dimension will become trivial foreign but and some things that would not that do not really appear per see in the set space will become relevant so they will become part of the state space as experienced by the agent but the hard part is to account for how this takes place um as a students by the agent becomes this at space per C how you have Matrix of 6 or 6 that serves as a bandarita to say and how agents will basically collaboratively build systems meaning where they have alphabet and can communicate long sentences using this this grid this is not in and of itself a question of structural learning although solving City learning wasn't able to to have a lot of computational experiments that would help greatly with uh looking at the fine grain structure the question is how you use how you use mathematics in a way um well in a way of describes creation this has the the construction set spaces but that's what I'm suggesting space that's what I'm suggesting when um so the you know what what I'm talking about is if if you could figure out when to add a new node or connect or uh Edge to your generative model like precisely when is it appropriate to add a new parameter when is it appropriate to say oh these two parameters are connected to that space in these proportions if you had that worked out I I could see a kind of straightforward resolution to what you're talking about I'm not saying like you know there is the system just unfolds dynamically and occasionally you know it's it learns new structure or parameters which basically means like you know the new degrees of freedom are added to the system you are again talking at the epistemic level you can't describe a lot of things um by basically looking at the Dynamics and being smart about it you don't you don't even need uh advanced mathematics to do that this statement we make is different let us say that you have people that agree that shells are an appropriate means of exchange of our x y and z uh then money becomes a thing it is a thing that exists individually because uh it has structural properties and these properties drive observatables through Dynamics equally individually did not exist one before the question we ask is how we account for an evolution within a space that builds another space we cannot just look at between the space and declare how it has changed it is not the problem we are posing problem is how it's reason mathematically something that occurs within space that produce another space that is the same but not the same that is structurally um similar that inherits from what existed before but is different in Collective aspect because this one thing became so what we have to have is the theory of becoming basically we can be very smart about modeling without having a theory of becoming we can be very precise in chemical inference and very efficient at seeing regime changes that does not mean we have a self-consistent theory of becoming creation space construction correct what we like those are two different problems and so one may help the other greatly both may help the other greatly because um the thing we are looking at is the same word looking at two very different aspects of the thing Maxwell want to add something or Alexa on that no I'm gonna think about that um thank you Emma awesome yeah from the chat on that topic Tom wrote a similar question Avila could you elaborate more on how active could be expanded to accommodate State space creation assuming active is normative about free energy minimization when should an agent expand the state space and Alex Kiefer also kind of uh added a note it's appropriate to add a new parameter when the model complexity accuracy trade-off says so more seriously isn't free energy measurable across models or different state spaces so it's almost like there's the structure learning meta cognition layer and it almost is like can the Box be wrapped within the Box can it be possible to have a state space that within its state space represents itself and or to what extent do these physical intermediates and their Associated physics and metaphysics need to come into play for different systems and at what level of generality basically about this I'd subscribed to a variant of consistent history interpretation of quantum physics but it's not quantum physics is just something that has the kind of ethical parties quantum physics has well I did commit to Quantum information several years so okay let us say it is a in nature Quantum which is a bit trivial because yes it has PCS in it which is what being Quantum is but whatever this is the most specific uh consistent way to presented from you still see those slides yep EBA EBA e and this is basically category Theory because this diagram as a property which is that uh following the arrows basically change what the space is B begins as a neutral interface so a space of possibilities that in principle are for interaction but it is not semantically loaded and the fact that you have the integration of e the observatables and a I from data as an agent but it would rather be agentive capabilities or constraints that enable observation of E this does not conserve the property of B Because B becomes equipped with a semantics so real theories that look at spaces without looking at what is in spaces those are category series and those are the most abstract field of mathematics as far as I know and I do not afford uh semantically loaded things like Furniture is a measure of not only it entails the states but it also entails a measure of States I think the basic mathematical structure we need to represent that is much more basic than any other things and any energy measure will break down much before we get there but we are that is more expensively loaded is this The Duality between the experience of struggle to landscape by agents and the introduce flow that is directed by the landscape and reconstructed landscape the question was what was the question could you repeat it word forward Daniel it was just about the uh normativity around model selection and whether modeling State spaces could or would require artifact the innovativity um is uh there it is in this representation of the problem uh I have a small agent that represents uh circular landscape and acts within it and I can't just claim okay there is that space there is a Flawless at space I can defend for energy Resurgence minimize strategy and then this is where structural learning intervenes basically because those agents will try to understand the ones they do not experience the world directly they will infer the world in a way that is robust and competent but they can make up new things they can fail to understand they can cheat they can erode the structure of the landscape that is collectively enacted in many ways and this again is dual to something that is the uh chemical it's not dynamical actually I do not know what it is it doesn't work the uh unfolding of uh the structure of the landscape under its own Sound flow that again is not dynamical because it's not what we call sensory size so we described what you're talking about in terms of Niche Construction yeah so uh in human systems I thought yeah we had a Carl and I had an interesting discussion about this a few weeks ago and uh yeah I think what what really distinguishes human systems from non-human systems is the degree to which the active inference Loop is mediated by uh what would otherwise be internal States but that end up having this kind of ambiguous function of yeah they're environmental that they they encode the material aspects of some normative practice uh or some they store some information uh about what you're doing like a like a a Blackboard for instance or or these slides that we're all looking at and you know in some sense encode the the thought or aspects of the thought that you're trying to communicate um so the yeah maybe a lot of the problem here can be finessed by noting that it's not just about agents interacting with each other there is a kind of common store of you know or traces you know like a kind of material thing that we all have in common yeah you have a common uh collective in action of a given structure to landscape because humans are such that then they synchronize in a robust and sponsored loaded way and that Circle the landscape it does not exist it is not something that pre-exists the fact that agents do that agents infer properties and then enact them and this make it an open landscape and the normativity of the fep affords probably a computational experiment design that help us see how such landscape unfolds and look at Material Theory where the agent is absent where it is landscape that produce landscape and once we have that we have a theory of creation it is what the theory of creation is so so what you're saying is if we had a theory where the agent was able to like expand its state space through structure and parameter learning in particular then we would have what you're looking for am I misunderstanding uh yes but I'm not sure that we agree that we're aligned on the constraints of these segments entails because we the goal is not to understand how the agent engage with personal environments it is to abstract if we get uh to the Target we can abstract entirely the agent in what they think and just look at constraints and how they informed I went to light on the constraint technology because we do not have it was not the place for that but we need to look at specifically constraints over what is observable and how they are invigorated and how they produce other constraints etc etc we have to get back at foreign I mean you're talking about the same thing so but the formulation uh that is uh taken in uh physics Often by belief is DST it entails uh promptive space and this is what needs to be um to address the problem I want to address this is why to have to be a overcome and this is likely possible given a constraints and trick ontology so ontology where do not have possibilities then constraints about the possibilities but constraints then possibilities afforded by and I do know what I'm saying is that those two stories are the same story uh that's a Bayesian mechanics cells that that yeah it is from within DST uh but they're both from DST what the I mean so a uh free energy principle theoretic model of the belief the updating of a particle in a system is is in the strong sense of like is dual to you if I have one I have the other right is uh a maximum entropy dynamical systems model of uh the whole system as in the whole system of particles and environment to which it's coupled and all of that sensory motor like that that's okay that's the cool thing it's the the the the fep is the canonically optimal dynamical systems model uh it's maximum entropy in the sense that you know it uh is the it makes the least assumptions given your current state of knowledge about the system you know it's James uh James as in you know j a y and yes it's Jane's optimal in the parlance of daltonville uh yes so if you have a constraints model you also have an fep model and vice versa if you work it out from the perspective of the FTP you can map back onto a constraint but definitionally constraint model with no reference whatsoever to the micro System Theory or pre-existing species so yes I agree you absolved the problem but the paper which referred to frames it within DST which is those that we need to get Beyond to solve this specific problem again I did not mean to say basically you have a you have uh framed this equivalence I think up to details let us have two details I would need to think about it actually um but uh with hypothesis that are precisely those that I want to use the equivalence to overcome you know constraints don't get you out of the problem that you were posing originally either like both no in you just you have another kind of layer of you know pre-given is not we have to characterize constraints it is that we have to have an ontology of constraints so much medical constructs that use this constraints with no reference to anything else just constraints questions I don't know but if we have that in the sense while conserving the sense that these are given by Mercy Moreno and while leveraging the equivalence between constraints and belief that well meaning belief whatever that is derived we have one but it is not something that has been done because fep is friend within uh well one eiffp vfcp is a friend within DST which press release States and qfvp is one does not specify what buildups library just specify that there are observables that we can make survivals but we can't solve this within DST or within a pure abstract Quantum function series is not afforded by the structure of these theories once because it states the other principles of servers we have to show how those are become integrated and the only way forward that I see is a constraint ontology so right so what I'm saying is this is mathematically speaking if you're talking about constraints then there is a in the end it's probably functorial I mean you know there is a rigorous and systematic mapping from what you're talking about to what the FTP is already saying um that's that's really yeah I mean that that's that's really what The Duality is saying like yeah uh maximizing your entropy with respect to some set of constraints is dual to minimizing your free energy with respect to some model it's the same thing if you have one you always have the other if uh how would you frame the notion of a constraint in your sense of Monsieur montervier and Moreno with no reference to either GST or in related States well I mean in these reaction networks that they're mapping they are implicitly using some kind of space production networks it is a another line of work right but like I'm just saying that there is a similar kind of like individuation of things that's going on when you map out like some kind of constrained uh schematic in in that sense like yeah the uh the point that you're making sort of like grandfathers both here which is okay but how do you you know write down a state space model at all and how do they how do you grow them in particular um but I guess I'm just saying like yeah constraints are just the subject to this as to the extent that the fep is subject to your criticism the constraint stuff is also for the same or dual reasons like if you just inverted all of the arrows in your reasoning you would also be able to apply this to the constraint stuff equally like that's not an exit to what you're pointing out but I guess my my complimentation if you think there is a uh if you think moving to a constraint ontology helps us then that's interesting because any like constraint mapping is also dual to an fep theoretic model uh so maybe the fep can be used like very directly just based on the way that you're like if you solve the problem in terms of constraints you could flip it back it doesn't really uh and it I don't know if you're stuck along the way then you can flip it at that point and think okay well does looking at does looking at it from this like do I want to look at self-organization from the perspective of the selves that are self-organizing are from the perspective of organization and you know the maximization of entropy and all that stuff I guess so if you're arguing that the FTP treats an entire different people naturally different problem this is not of your hold it is a viewer opposed specifically I do agree that physics of and by belief derives within the St rigorous relation between basically what are the two sides of this up to details um what I am saying is framing as a goal the fact to leverage that other way to of space construction and I also claim that this would be essentially equivalent uh to build a self-consistent and automatically consistent constraint Anthology for uh for for that for this diagram um I'm not sure what you're going to get because it is an ambiguous that your research has of noise framed within DST which precipices states which the constraint topology would not and it is indigo that he is very relevant because it's as you said entails one Duality at least in one specific case which is list of DST that is uh the key problem to address so I think you wait you may be combining two issues in this worry I mean so that there is the worry of like new States arising and there's the worry of using dates at all so I mean I think yeah so it's the problem is not really that you know dynamical systems theory assumes States it's just the static nest of the or the you know the pre-determinedness or the priests best defiantness okay so yeah so specifically there is a very fruitful Way Forward I think to address a lot of them my uh my proposal is a state space construction not the state space stationism to get uh uh right I need B to begin with I need physical things to be there well to have a actually self-consensory you would need this to work from uh X nilo to get something from nothing but it is not the issue attend here the issue at hand here is how existing things can unfold create whatever meaning States so it is not the notion of states that is the problem the problem is uh the fact that we do not have a theory of how states are created and we cannot frame it exclusively from uh mathematical survey that by construction precipose is a space of from this Theory to generalize it to the space construction which is what I'm proposing so but what about the renormalization group so if you start really from the idea that you just have you know observations and they uh you know have a scale of noise associated with them um yeah you can I think I mean there are principled you know physics the kind of ways of bootstrapping something kind of saying well you know some stuff changes so fast relative to my reference scale that it basically averages out and just contributes kind of random fluctuations to what I'm observing and you look at basically why to personalize it to look at relevant scales but maybe it would be afforded to have a hospitalism but it does not entail it does not know the other password realism it is a model of how complex things complex Dynamic unfold so it is relevant it is not what I'm pointing at right as we come near the end for each of us maybe you can spend the most time at Apple but for each of us what are our next steps what cognitive counterfactuals and physical embodiments are going to be shaped by this work and maybe we can have non-avail go first and then you can have the last word sound good okay well personally I'm very excited to work on the category Theory this year learning about the recent uh dissertation of uh Toby Smith also I think it's interesting to bring in that Niche modification active inference variation ecology a lot of the work that as Maxwell brought up earlier had been framed within a multi-scale setting bringing some of these new developments over the last one or two years like the path-based formalization renormalization group environmental process uh generative process generative model symmetries weak Markov blankets there's just many new Concepts and toolkits on the table today in 2023 then I think there were one or two or three years ago so it's been a super interesting discussion and it brings a lot of topics to our attention that I expect will be able to produce her uncertainty about a lot in the coming year and I'll pass to Alexi then Maxwell and then Apple the opposite it would be more consistent because Alexi and I work directly together in Kairos okay Maxwell um well I mean Daniel I would I would Echo your overall uh assessment I share your enthusiasm for where like everything is at right now I think we're poised for some very exciting work and a lot of very uh interesting stuff has happened over the short term uh in terms of what others presented um I think uh you know uh in addition to what you were describing one of the things I want to uh like work out this year is really um the philosophy of the free energy or Bayesian mechanics I think uh up until recently what what most everyone has been doing myself included was just coming at this thing from whatever you know disciplinary background we had access to in trying to just make sense of it but uh I think that has made us kind of blind to some of the and everyone has kind of their blind spots you know all of these so-called interpretations of the fep signal to me that there's there's actually yeah there's there's philosophical work to be done to kind of understand uh Jules Hedges put it nicely on Twitter I think he said that he's become convinced of two things first the fep is true in the sense that it's quote unquote the right way to do it and second that it's a lot more like boring than all the hype suggests in the sense that it's yeah it just follows from the way that a bunch of physics hooks up it's very very fundamental and important but like yeah I mean um there are it it's it's part of the kind of core package now and there are new exciting Horizons of like unknown that you know more more to the the point is the the there's there's like a a kind of core philosophy to the fep that remains to be uh developed so for example The Duality between the principle of Maximum and should be uh which you know implements a form of like Occam's razor in some sense and the free energy principle I think is kind of deep and that we've only started to think about you know the implications that happened Chris fields's group has been suggesting that there's a similar Duality between the fep and uh unitarity or the conservation of classical information in um yeah Quantum information Theory so uh that's very exciting sorry for talking so long awesome exciting times indeed all right Alexi and aval with the final words yeah so uh I will be shot so I'm really excited about how these debates will move the lines in theoretical but also practical philosophy and on our direct perception of staphy life in general to be a bit fraud uh also I hope that part of this will make like I hope this will make the difference in metaphysics and Anthology and philosophy and methodontology also and naturalism of course as it seems to do in active Insurance lab and Kairos so we are three things we need to do to um settle the dust on the ornament I generated there the first is to formalize much better segment and ontological consistency and pastoralism both need to be very clear in their reach and entailments this is something we'll do with uh Alexi mainly and possibly Auto Group members something that we need to do is to put a lot of work in working out what this basis mean I do not know I can tell you that it is essentially correct I'm pretty confident that's essentially correct but I have no idea whatsoever what it means click which is pretty big problem as far as I'm concerned so there will be a lot of work uh for me and other members namely yunabrina can the maximum to work out the specifics and of this of this mathematical column thing that allows order to unfold from most pressingly in my opinion at least it's it's what I'm supposed to focus on in terms of my structural flow it is to work out what this means how social constraints are enacted and how they evolve because ultimately the rest is abstraction this is what has content this is what allows us to say okay these are what exists these are the constraints that shape things and this is what is their flow um it is critical from scientific perspective because it is what gives me to to listen to her accounts it is critical from a well pragmatic perspective because it is uh what tells us what world we are in and what is its flow which is the goal of Kairos something we'll do with um uh namely manual job and uh um so basically we will work around the specific of this and we will uh if someone decides you to collaborate please write us we will be very interested uh because there is a lot of all of these problems are big problems all in state a lot of Firepower so yeah basically anyone who wants to who's interested in working on the specifics with us he's very much invited to reach out excellent thank you Alexi apple and Maxwell for joining thanks everybody for watching till 34.2 all right it's over thanks yeah thanks both fellas um